

A `for` loop in Python is used to **iterate over sequences** like lists, strings, tuples, dictionaries, and ranges.

1 Basic `for` Loop Syntax

```
python

for variable in sequence:
    # Code block to execute
```

2 Iterating Over a List

```
python

fruits = ["apple", "banana", "cherry"]
for fruit in fruits:
    print(fruit)
```

Output:

```
nginx

apple
banana
cherry
```

👉 Loops through each element in the list and prints it.

3 Using `range()` in `for` Loops

```
python

for i in range(5):
    print(i)
```

Output:

```
0
1
2
3
4
```

👉 `range(5)` generates numbers from `0` to `4`.

Custom Start & Step

```
python

for i in range(1, 10, 2): # Start=1, End=10 (exclusive), Step=2
    print(i)
```

Output:

```
1
3
5
7
9
```

4 Iterating Over a String

```
python

for char in "Python":
    print(char)
```

Output:

```
css

P
y
t
h
o
n
```

👉 Loops through each character in the string.

5 Using `else` with `for`

```
python

for i in range(3):
    print(i)
else:
    print("Loop finished!")
```

Output:

```
vbnet

0
1
2
Loop finished!
```

👉 The `else` block runs **only if the loop completes without a `break`**.

6 Breaking the Loop (`break`)

```
python

for i in range(5):
```

```
        break # Stops the loop
    print(i)
```

Output:

```
0
1
2
```

👉 `break` exits the loop when `i == 3`.

7 Skipping an Iteration (`continue`)

```
python

for i in range(5):
    if i == 2:
        continue # Skips `i == 2`
    print(i)
```

Output:

```
0
1
3
4
```

👉 `continue` skips the rest of the loop for `i == 2`.

8 Nested `for` Loops

```
python

for i in range(3):
    for j in range(2):
        print(f"i={i}, j={j}")
```

Output:

```
ini

i=0, j=0
i=0, j=1
i=1, j=0
i=1, j=1
i=2, j=0
i=2, j=1
```

👉 Loops inside loops!

9 Looping Over a Dictionary

python

```
student = {"name": "Raj", "age": 21, "city": "Chennai"}
for key, value in student.items():
    print(key, ":", value)
```

Output:

yaml

```
name : Raj
age : 21
city : Chennai
```

👉 Loops through key-value pairs.

10 List Comprehension (Shortcut for `for` Loops)

python

```
numbers = [i * 2 for i in range(5)]
print(numbers)
```

Output:

csharp

```
[0, 2, 4, 6, 8]
```

👉 Creates a list using a `for` loop in one line.

🔥 Summary

Feature	Example
Basic `for` Loop	`for x in list: print(x)`
Using `range()`	`for i in range(5): print(i)`
Looping Over String	`for char in "hello": print(char)`
`break` (Stop Loop)	`if x == 3: break`
`continue` (Skip Iteration)	`if x == 3: continue`
Nested Loops	`for i in range(3): for j in range(2): print(i, j)`
Looping Over Dictionary	`for k, v in dict.items(): print(k, v)`
List Comprehension	`[x * 2 for x in range(5)]`

What Else Do You Want to Learn About `for` Loops? 😊